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Review

Balancing watershed and local scale impacts of rain water harvesting in India—A review

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ABSTRACT

Agricultural production in India has become increasingly reliant on groundwater and this has resulted in depletion of groundwater resources. Rainwater harvesting (RWH) for groundwater recharge is seen as one of the solutions to solve the groundwater problem. This is reflected in an increase in watershed development programs, in which RWH is an important structural component. Understanding the net effect of these development programs is crucial to ensure that net effect on groundwater is positive both locally and within a watershed. Hence, this review focuses on the hydrological impacts of RWH for recharge at the local (individual structure) and watershed scale in rural areas. Surprisingly little field evidence of the stated positive impacts at the local scale is available, and there are several potential negative impacts at the watershed scale. The watershed scale is underrepresented in the field studies and is mainly approached through modelling. Modelling is seen as a possible tool to extend limited field data and scenario studies can be used to examine potential impacts. However, many past modelling studies examining RWH have either had limited focus or have been based on insufficient data. Development of new modelling tools is needed in combination with increased field data collection. Increased use of remote sensing and advanced statistical techniques are suggested as possible new opportunities. In addition, some evaluation criteria are proposed to assess the local and watershed scale hydrological, and other, impacts of RWH as part of watershed development.

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1. Introduction

In India, groundwater accounts for more than 45% of the total irrigation supply (Kumar et al., 2005), and for about 9% of India's Gross Domestic Product (GDP) (Mudrakartha, 2007). This has not always been the case; over the last 50 years India has seen a huge boom in the use of groundwater, resulting in an exponential increase in the number of tube wells to an estimated total of 19 million in 2000 (Shah et al., 2003). As a result agricultural livelihoods of small-holder farmers in India have improved dramatically because groundwater requires little transport, can be accessed relatively easily and cheaply, is produced where it is needed and provides a relatively reliable source of water (Dhawan, 1995). However, it has also contributed to serious groundwater depletion, with the water table declining at the rate of 1–2 m/year in many parts of India (Rodell et al., 2009; Singh and Singh, 2002).

The main replenishment of groundwater is through recharge from rainfall, covering both diffuse sources (as leakage below the root zone of vegetation) and focussed sources (through transmission losses from rivers and from lakes and ponds) (de Vries and Simmers, 2002; Lange, 2005; Shentsis et al., 1999). Recharge can be highly variable and total volumes are difficult to predict (Bouwer, 2002). In India, this is exacerbated by the fact that rainfall patterns are monsoonal with approximately 75–90% of rainfall concentrated in the summer months, June to September (Mooley and Parthasarathy, 1984).

As a result of this rainfall pattern, India has a long history of rainwater harvesting (RWH) (Sakthivadivel, 2007; Shah, 2001). In many rural areas of India, a specific purpose of RWH is to catch and store monsoonal runoff, which then percolates to groundwater tables (Keller et al., 2000). Given the current threat of groundwater depletion and the potential of increasing recharge, the implementation and planning of RWH continues to grow in India (Agarwal and Narain, 1997; Shah et al., 2009). However, the economical value and long term sustainability of structures in terms of maintenance has been questioned (Bouma et al., 2007, 2011; Raju et al., 2009).

In practice, the impact of RWH on the hydrological balance of a watershed is that water is stored and delayed with a transfer of surface runoff into groundwater, evaporation and transpiration. This can also be understood as the transfer of 'blue' water (rivers and aquifers) to 'green' water (soil water and plant water use). As more water is 'captured' through irrigated land use, blue water is converted into green water (Falkenmark, 2003). The potential increase in available groundwater may encourage increased groundwater abstraction for crop irrigation or other uses resulting in socio-economic impacts, while the impact on the water balance may be zero or negative. Hence, in general, RWH will change the water balance within a watershed. From a watershed perspective, this means it is important to quantify the hydrological impact of RWH structures and the related downstream trade-offs for a given level of watershed development. To achieve this, the changes in the spatial and temporal distribution of water and the changes in the volume of blue and green water would need to be quantified.

Such quantification can be complex, because the local hydrological impact of RWH will depend on factors such as geological and geomorphological settings, RWH local watershed size, design of the structures and the nature of the underlying groundwater system (Mishra et al., 2010). As a result, some quantitative studies on RWH have focussed on identifying optimal sites for RWH in order to plan watershed development programs (De Winnaar et al., 2007; Jasrotia et al., 2009; Kahinda et al., 2008; Mbilinyi et al., 2007, 2005). Overall, this research is fairly applied and mostly based on remotely sensed data. More importantly, for the overall watershed scale, many other factors need to be considered, beyond the questions of where to install RWH and how many structures can be built in a single watershed. For example, this might need to include the spatial distribution of RWH structures relative to the spatial variability of rainfall, in combination with the distribution and management of groundwater demand. Finally an assessment of the overall groundwater sustainability would be needed.

Definitions for groundwater sustainability are argued in many papers (Kalf and Woolley, 2005; Loucks, 2000; Sophocleous, 2000), and it is often defined as safe yield, or the maintenance of a long-term balance between the annual groundwater withdrawal relative to the recharge (Sanford, 2002; Sophocleous, 1997). More recently, several authors have argued that this is too simplistic (Alley and Leake, 2004; Sophocleous, 1997, 2000) as it does not take into account capture, which is the reduction in groundwater discharge or increase in recharge (Kalf and Woolley, 2005; Maddock and Vionnet, 1998). Hence, understanding the impacts of demand and supply-side groundwater management, including extraction for irrigation and recharge from RWH, are important to understand and enhance groundwater sustainability.

It has been suggested that despite the widespread use of RWH techniques for groundwater recharge in India, there is limited research examining the combined local and watershed scale hydrological impacts of RWH, and this limits socio-economical analysis beyond the traditional cost–benefit analysis at the local scale (Machiwal et al., 2004; Srivastava et al., 2009). For example, Bouma et al. (2011) used annual totals for hydrological variables in their economic analysis of the watershed scale effect of rain water harvesting, which ignores dynamic seasonal effects. Also, there is a need to consider meso-catchment scale impacts of RWH, particularly in rainfed agriculture, where RWH plays an integral role (Rockström et al., 2010). In addition, we could not find any comprehensive review of RWH with particular focus on groundwater augmentation in rural India.

This review therefore examines the existing literature on RWH hydrological impacts on groundwater systems and watershed scale water balances to understand the knowledge gaps. These knowledge gaps are important to identify because the goal of RWH is to have long-term positive impacts on people's livelihoods, which needs to be achieved without major environmental impacts. This review will specifically focus on RWH for groundwater recharge in rural areas, which is where most of the watershed development in India takes place. The paper first reviews the groundwater problem in India and the definitions of RWH. It subsequently reviews local scale methods and studies that measure the hydrological impacts of RWH before evaluating how this affects the overall watershed scale. Finally, this review further aims to develop a set of evaluation criteria to assess the hydrological and other impacts of RWH at the local and watershed scales. This can assist with the planning of watershed development of RWH structures and the development of policy to guide further investment in RWH vis à vis groundwater sustainability in India.

2. Background

2.1. Groundwater use and problems in India

Eighty percent of global groundwater use occurs in Bangladesh, China, India, Iran, Pakistan and the US (Shah et al., 2007), with India being the largest groundwater irrigator in the world (Shah et al., 2006). Groundwater development has been extremely important for rural poverty alleviation. In India and China combined, 1–1.2 billion poor small-holder farmers are supported by groundwater (Shah et al., 2007). This is because groundwater irrigation tends to be less biased against the poor compared with large scale surface water irrigation projects (Deb Roy and Shah, 2002). Groundwater is

easily accessible and can be developed quickly by farmers or small groups, and is reliable and flexible in time and space. Furthermore, from a water management perspective, groundwater also has less evaporation losses than surface dams or canals (Keller et al., 2000). In India, groundwater-based irrigation covers a greater area than the established canal irrigated systems, which were set up largely by the colonial government in the late 19th century (Sakthivadivel, 2007; Shah, 2007).

In India, the technological advance from shallow wells, with animal pulling and human labour, to diesel and electric pumps has greatly impacted the amount of groundwater extraction (Shah et al., 2006). While the growth in groundwater use over the past few decades has improved rural livelihoods, there are increasingly serious issues with aquifer depletion (Shah et al., 2007). Currently the response to groundwater depletion has focussed mainly on the supply-side management. In India, a massive integrated watershed development program provides public resources to local communities, for works including constructing RWH structures (Hope, 2007; Shah, 2007). Methods to recharge aquifers, including RWH (sometimes referred to as artificial recharge), have become so widespread in India over the last two to three decades that it is now referred to as 'a groundwater movement' or 'artificial recharge movement' (Sakthivadivel, 2007, 2008). However one of the difficulties of assessing RWH hydrological impact is the lack of evaluation of the impacts of these projects on groundwater, and any upstream–downstream trade-offs (Hope, 2007; Sakthivadivel, 2008). In fact, Kumar et al. (2006) suggest that the reliance and depletion of groundwater in India has caused investment in RWH to occur rapidly without any hydrological assessment.

A further complication is caused by the occurrence of salinity in some of the aquifers in India (Misra and Mishra, 2007; Rao et al., 1997; Subba Rao, 2006; Subramani et al., 2005), but we could not find a consistent overview of groundwater salinity across India. A particular issue is the occurrence of saline groundwater in deeper layers, with fresh groundwater floating on top. Continued pumping can cause rapid depletion and increased mixing of the two layers (Asghar et al., 2002; Kamra et al., 2002) resulting in an overall degradation of water quality in the system.

2.2. Definition and general overview of rainwater harvesting (RWH)

Rainwater harvesting involves using small-scale structures to collect runoff for either domestic use (Evans et al., 2006; Handia et al., 2003; Mwenge Kahinda et al., 2007; Villarreal and Dixon, 2005), supplemental irrigation as is most common in Africa (Ngigi et al., 2007), or for groundwater recharge, as is typical in many regions of India (Kumar et al., 2006).

RWH for groundwater recharge can also be described as artificial recharge (Bouwer, 2002). However, in much of the literature RWH is considered part of 'managed aquifer recharge', which includes artificial recharge, enhanced recharge, water banking and sustainable underground storage (Dillon, 2005; Gale, 2005). Managed aquifer recharge is defined as the 'planned, human activity of augmenting the amount of groundwater available through works designed to increase the natural replenishment or percolation of surface waters into the groundwater aquifers, resulting in the corresponding increase in the amount of groundwater available for abstraction' (IETC, 1998). This is the definition that we will use in this review.

More generally, RWH encompasses methods to induce, collect, conserve, and store runoff from various sources and purposes, by linking a runoff-producing area with a separate runoff-receiving area (Boers and Benasher, 1982; Rockström, 2000; Young et al., 2002) (Fig. 1). Methods of RWH have three common characteristics (Boers and Benasher, 1982):

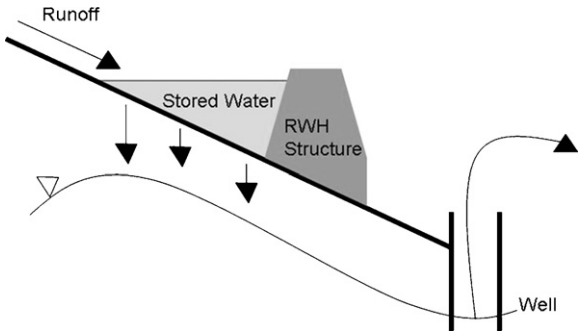


Fig. 1. Schematic representation of RWH function for groundwater recharge. Water from runoff is stored in the structure and allowed to infiltrate. The infiltrated water can subsequently be extracted via local wells.

1. RWH depends upon small-scale capture of local rainfall and/or runoff. RWH does not include storing river water in large reservoirs or the mining of groundwater.
2. RWH can be applied in arid and semi-arid regions, where runoff has an intermittent character and rainfall is highly variable, so drought and flood hazards to agriculture are significant. In these areas storage of water is important. This means RWH is viable in areas with annual rainfall as low as 300 mm (Kutch, 1982), but this may also be a disadvantage, because RWH is dependent on limited and uncertain rainfall (Qadir et al., 2007).
3. RWH is a relatively small-scale operation in terms of watershed area, volume of storage, and capital investment, ranging from household, to field or small watershed (Ngigi, 2003). Mbilinyi et al. (2005) classified RWH on the size of the runoff-producing area:
   (a) On-farm, within-field systems or in situ. Rainfall is captured where it falls, to conserve water and prevent runoff from cropped areas and prolong the time for infiltration (Vohland and Barry, 2009).
   (b) Micro-watershed system, where there is a distinct division between the runoff-generating watershed area and a cultivated basin or storage area where the runoff is concentrated and stored (Gowing et al., 1999). In micro-watershed water harvesting, the percentage of runoff increases with decreasing watershed size due to reduced infiltration losses. Small watersheds can produce 10–15% of annual rainfall as runoff (Boers and Benasher, 1982).
   (c) Macro-watershed system, which has a larger watershed area than a micro-watershed system.

From as early as 4500 B.C., RWH has been practised in various parts of the world (Verma and Tiwari, 1995), and is most commonly found in developing countries due to its decentralised, low cost and local-scale aspects. In India, RWH has been practised for at least the last 1000 years (Agarwal and Narain, 1997; Pandey et al., 2003; Sakthivadivel, 2007; Shah, 2001). Traditional RWH systems were practised in the Indus valley as early as 3000–1500 B.C. (Grey and Sharma, 2005). Techniques of RWH have been noted in ancient texts, such as the Rigveda (1500 B.C.) and the Atharva Veda (800 B.C.) (Agarwal and Narain, 1997). According to Pandey et al. (2003), abrupt climate fluctuations heightened construction efforts of RWH structures in prehistoric and early historic societies. Despite the long tradition of RWH, it became neglected from the time of British rule (Radhakrishna, 2003). With Indian



Fig. 2. Example of a RWH structure known as an *Anicut* in Rajasthan, India. At the end of the monsoon in September the structure is full. Three months later the storage is almost empty, through evaporative loss, lateral subsurface flow and recharge.

independence, the state became the major provider of water, replacing communities and households as the primary units for provision and management of water (Agarwal and Narain, 1999). In the last few decades, RWH has seen a strong revival due to groundwater depletion, involving the participation of communities, government and non-government organisations (NGOs). It has been estimated that in India today, almost 1.5 million traditional village tanks, ponds and earthen embankments harvest rainwater in 660 000 villages (Pandey et al., 2003).

In India, while somewhat depending on the region and type of aquifer (for example alluvial or fractured hard-rock), the main purpose of RWH is to store runoff to recharge shallow groundwater aquifers (Agarwal and Narain, 1997) (Fig. 1). While developed as a result of the concentration of rainfall in the monsoon from June to September of each year, methods are highly location specific (Verma and Tiwari, 1995). The normal duration of the monsoon in India is about 100–120 days and the Indian plains receive about 80% of annual rainfall during this time (Kumar et al., 2005; Ramesh and Yadava, 2005), thus RWH stores runoff that could otherwise continue downstream. Depending on the geology, stored water can percolate into the underlying groundwater table (Fig. 2), which is subsequently used for irrigation and domestic purposes via dug wells or tube wells.

However, RWH generally consists of open storages which can be subject to high evaporation losses, due to high surface area to volume ratios (Neumann et al., 2004). Thus if the infiltration rate of the stored water is slow, most of the water will be lost via evaporation. Conversely, storage in aquifers has the advantage of essentially zero evaporation (Bouwer, 2002). Rainwater harvesting also tends to lead to increased crop production intensities and greater crop yield, because rises in the water table mean better accessibility and yields of groundwater (Keller et al., 2000). This feedback between RWH and irrigation area is an important consideration due to the impact on groundwater sustainability (Glendenning and Vervoort, 2011). Siltation of structures is an issue for all dams, even with controlled and minimum erosion. For RWH structures to remain functional and to maximise recharge, they need regular maintenance through clearing of accumulated silt (Gunnell and Krishnamurthy, 2003). However, maintenance of RWH structures is attracting only limited attention in the reviewed literature. Local RWH management may be inspired through the Hindu tradition of collective action (Cochran and Ray, 2009), but Bouma et al. (2007) suggests that there is little incentive for local people to contribute to long term maintenance and operation of RWH structures, a view also shared by Gunnell and Krishnamurthy (2003).

Even if the underlying groundwater is saline and unusable for irrigation or drinking water, RWH can provide local increases in fresh water as the density of fresh water is less than for saline water. As a result, fresh water lenses that "float" on top of the saline groundwater are recharged (Asghar et al., 2002; Cendón et al., 2010) and can be harvested via wells. The effectiveness of such a system depends on the level of mixing between the recharging water and the underlying saline groundwater which in turn depends on the recharge rate and the lateral transmissivity (Eeman et al., 2011). In addition, additional recharge of water in a saline aquifer can lead to increased expressions of salinity at discharge points (Kennett-Smith et al., 1994).

3. Methods for measuring the hydrological impact of RWH for groundwater recharge

3.1. Field measurements of recharge are difficult but important

Influencing recharge is the key hydrological impact for which RWH is designed. However, RWH impacts need to be assessed at different scales. At the local scale the impact should focus on the amount of recharge that is produced by each of the structures, and how much of this water is available for subsequent irrigation and production. One reason for the lack of studies on the impact of RWH is because measuring the average recharge of many RWH structures across a watershed can be difficult, expensive and time consuming due to high spatial and temporal variability in landscape and climatic conditions.

As Section 3.2 will show, many field studies of RWH impact are based on calculating recharge using the water balance. However, recharge is one of the most difficult components of the water balance to measure, because it needs to be measured below the visible surface and is highly variable. Particularly in arid environments, where it tends to be the smallest component of the water balance (Bond, 1998).

More problematically, all well-established recharge-estimation methods have limitations, most of which yield results that are problem and scale dependent (de Vries and Simmers, 2002; Healey, 2010). Detailed overviews of these measures have been given in several other papers and books (Bond, 1998; de Vries and Simmers, 2002; Healey, 2010; Scanlon, 2004; Scanlon et al., 2002) and will not be repeated in detail here.

Several new recharge measurement tools have been developed which may aid in reducing the cost and time required for detailed field studies. The first is the application of remote sensing. Becker (2006) gives a broad review of the different remote sensing techniques for measuring groundwater recharge. The strength of using satellite data is that it can be used for regions where data is sparse or non-existent. It can cover larger areas and thus potentially give an idea of the effects of RWH at the larger watershed scale. However, temporal and spatial coverage and the length of the records can be significant limitations (Becker, 2006; Galloway, 2006). A second new recharge measurement is the application of heat as a tracer (Constantz et al., 2002, 2003b). The concept is similar to that of using other tracers such as bromine or chlorine (Constantz et al., 2003a; Cox et al., 2007). For a thorough review on the various techniques for using heat as a tracer the reader is referred to Constantz (2008).

3.2. Examples of field studies measuring hydrological impacts of RWH for groundwater recharge

Given the expense and difficulties of field estimation of recharge, it is no surprise that so far field studies to quantify recharge from RWH in rural regions have been limited. Of the field studies reviewed, the preferred method for measuring RWH recharge has been based on specific yield, such as the water table fluctuation method, coupled with a water balance approach (Glendenning and Vervoort, 2010; Sharda et al., 2006) or using water level well measurements (Badiger et al., 2002; Gontia and Sikarwar, 2005). Other methods have focussed on groundwater tracing using isotope and geochemistry analysis to assess the contribution of artificial recharge (Stiefel et al., 2009).

The results of the local scale field studies of RWH in India highlight the variability in the aquifer response to RWH and the complexities of taking a full range of measurements. Gontia and Sikarwar (2005) reported groundwater level rise of 8 m in the Saurashtra region of Gujarat and this rise was assumed to come from RWH, though no measurements were taken from the structures themselves, nor was there any comparison to background rainfall effects. Similarly, Raju et al. (2006) indicated rises up to 2 m in groundwater wells downstream of subsurface dams used for RWH in the Swarnamukhi River. Badiger et al. (2002) inferred that recharge from RWH was about 3–8% of rainfall from well measurements near RWH structures. Sharda et al. (2006) quantified recharge from a number of RWH structures in Gujarat, using the water balance and specific yield methods. They found that structures had a maximum recharge capacity due to the design of the RWH structure. Of the annual rainfall, they estimated that 11% could potentially recharge from a structure. Most recently, Glendenning and Vervoort (2010) indicated recharge efficiencies of rainwater harvesting structures of around 7% of total rainfall in the Arvari watershed in Rajasthan. Isotope and hydrogeochemistry studies on wells close to RWH structures have indicated that up to 75% of the groundwater could originate from RWH recharge, but this percentage varies depending on hydraulic connectivity of the wells (Stiefel et al., 2009; Sukhija et al., 2004, 1997).

It is problematic to compare these studies, not only because of differences in geographical and aquifer characteristics, but also because of differences in methods used. Difficulties exist in comparing the estimates from hydrogeochemistry data and the water level data as the hydrogeochemistry studies tend to be snapshot observations (and lose the temporal dimension). In contrast, water level data cannot account for sources of water. Depending on the location of the measurement both methods can overestimate the total amount of recharge if measurements are taken close to a structure (e.g. Glendenning and Vervoort, 2010) (Fig. 3).

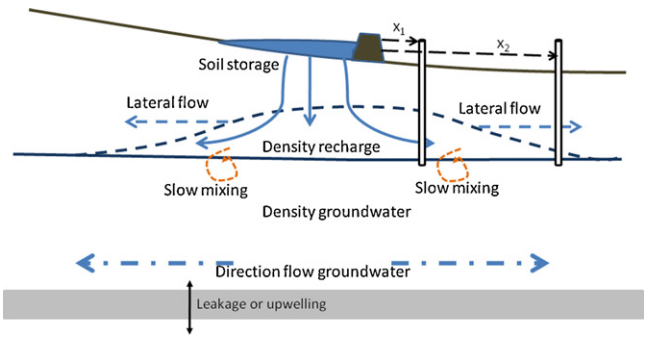


Fig. 3. Diagram indicating the possible complexities in measuring the hydrological impact of RWH in the field from an individual structure. This includes impacts of lateral flow, difference in density between infiltrating and recharge, leakage or upwelling to or from underlying aquifers and soil storage.

Furthermore, the hydrogeochemistry data does not distinguish between background recharge and focussed RWH recharge where both could have similar hydrogeochemical signatures. For exact quantification, all chemical sources of groundwater have to be taken into account (Dettinger, 1989), and sometimes not all of these are known. Differences in salinity between the recharging water and local groundwater could induce differential mixing and thus depending on the sampling location this could further overestimate the recharge contribution in hydrogeochemistry data (Fig. 3). This clearly highlights the need to develop a unified approach or mixed methods approach. The minimum requirement would be a clear comparison of different methods at a single location.

There is dearth of field studies looking at watershed scale impacts of RWH. In fact, we could not find any field studies that look at watershed scale hydrological impacts of RWH. Such studies would be difficult, both in terms of instrumentation to cover spatial and temporal variability and in the length of time needed for monitoring. Jain (2008) reports qualitative observations of flow in the Arvari river in Rajasthan post RWH watershed development by the local non-governmental organisation, suggesting an increase in flow duration. However, relating the reported months of flow in the river to annual rainfall at a nearby station suggests that climate impacts cannot be ignored (Fig. 4). This highlights the complexities of measuring the hydrological impact at the watershed scale. It also suggests that development of a strong quantitative project to monitor catchment scale flows would complement and potentially integrate other local scale work on hydrological impacts of RWH.

3.3. Using models to measure the watershed scale hydrological impact of RWH for groundwater recharge

Considering the difficulty, cost and time required for physical measurements of recharge, modelling often provides a cheaper, faster way to consider larger scale watershed hydrological impacts of RWH. A drawback of the modelling approach is that models are generally not better than the underlying data (Silberstein, 2006). Models range from general and/or analytical applications of the water balance to sophisticated three dimensional numerical models (Jha and Peiffer, 2005; Kahlown and Abdullah, 2004; Khazai and Spank, 1997; Sorman and Abdulrazzak, 1993). For example, MODFLOW has been used to successfully predict the impact of artificial recharge on streamflow (Barber et al., 2009).

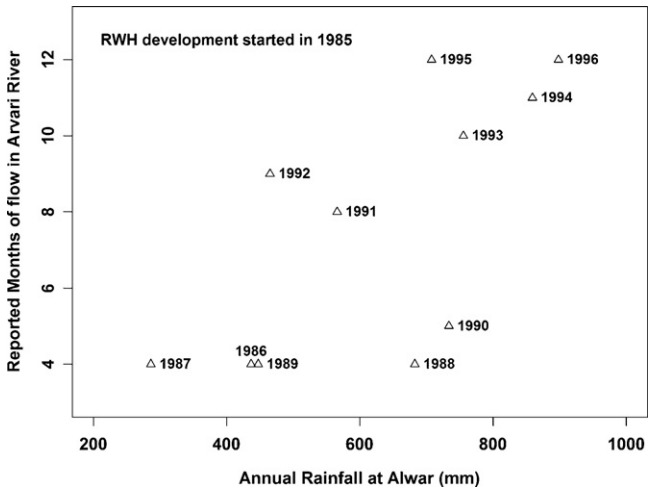


Fig. 4. Relationship between annual rainfall at Alwar and qualitatively reported months of flow in the Arvari river (Jain, 2008) in the years post implementation of RWH development. No information about the spatial location of the observations and no definition of "flow" is given in the source.

A number of modelling studies have looked at the amount of runoff that can be captured by RWH (i.e. the runoff potential) to prioritise watersheds for RWH development. Satellite images and Geographical Information Systems (GIS) are seen as a useful tool for this assessment (Anbazhagan et al., 2005; Sekar and Randhir, 2007; Sharma et al., 2001), which has led to a large number of papers using remote sensing and GIS for RWH suitability of sites (e.g. Chowdary et al., 2009; Jasrotia et al., 2009; Sharma and Thakur, 2007; Singh et al., 2009). Gupta et al. (1997) used the curve number method (Hawkins et al., 2009) combined with GIS to estimate runoff potential in a semi-arid watershed of Rajasthan for RWH development. Sharma and Thakur (2007) combine GIS and water balance calculations. They estimated that the shift in the water balance at the watershed level due to watershed development, including more RWH would result in a decrease of runoff by approximately 60% and an increase in recharge by only 5%. The main change in the water balance was a significant increase in actual evapotranspiration due to increased irrigation and due to changes in landuse. Few other studies have extended GIS analysis to consider potential RWH impacts on groundwater levels and surface flow or estimate how the number of structures would affect the watershed water balance.

In addition to the GIS approaches, both lumped and conceptual water balance models have been used to estimate groundwater recharge from different RWH techniques (Badarayani et al., 2005; Glendenning and Vervoort, 2011; Hassan and Bhutta, 1996; Ouessar et al., 2009, 2004; Pretorius et al., 2005). Specific to India, Gore et al. (1998) quantified the effects of RWH in 16 observation wells in Maharashtra state, by modelling groundwater coupled with a water balance model, and concluded that there was an overall increase in groundwater recharge from RWH of 16%, which is an increase of 2% of the local annual rainfall. Using the HYLUC (Hydrological Land Use Change) model in Andhra Pradesh, rainwater harvesting was suggested to cause the closure of a basin (Calder et al., 2008), which means no more water would flow further downstream. However, increased siltation in the RWH structures caused downstream flows to increase again. Glendenning and Vervoort (2011) developed a conceptual watershed water balance model, incorporating surface–groundwater interactions and hydrological response units. The model was based on field data from the Arvari watershed in Rajasthan to examine watershed-scale trade-offs and RWH. Using sustainability indices, they found that RWH increased viability of irrigated agriculture, though there are strong feedbacks between the two. Despite reducing streamflow, the model showed that RWH provided a slight buffer in drought years due to increased reliability of groundwater stores. Based on a theoretical modelling study, Neumann et al. (2004) concluded that the impact of RWH on water levels in a watershed may be minimal for all but the immediate vicinity of the structure itself.

Small farm dams are similar in function to RWH in India. Farm dam impacts have been assessed using simple lumped watershed models such as TEDI (Tool for Estimating Dam Impacts) (Nathan et al., 2005) and CHEAT (Complete Hydrological Evaluation of the Assumptions in TEDI) (Jordan et al., 2008; Lowe et al., 2005), generally indicating large reduction in flows. Using TEDI, Savadamuthu (2002) found that the impact of farm dams on capturing annual runoff is very high during drier years and is marginal during wetter years, which may lead to a non-linear type response (Callow and Smettem, 2009). The cumulative effect of a large number of storages within a watershed changed the spatial and temporal distribution of available water which caused dieback in certain downstream tree species in South Africa (O'Connor, 2001). Ouessar et al. (2009) adapted the complex watershed scale model Soil Water Assessment Tool (SWAT) to assess the impact of water-harvesting systems in a 270 km^2 arid watershed in Tunisia. However, they concluded that model had several deficiencies in the routing and groundwater routine. Sakthivadivel et al. (1997) used the model ROSES (Reservoir Operation Simulation Extended System) to model daily hydrologic behaviour of cascading small tanks to guide in small tank rehabilitation works in Sri Lanka. Letcher et al. (2001) looked at two algorithms for analysing changes in stream flow response due to changes in land use and farm dam development in three watersheds and Schreider et al. (2002) used the model IHACRES (Identification of unit Hydrographs And Component flows from Rainfall, Evaporation and Streamflow) to see statistically significant reductions in the quantity of potential streamflow response caused by farm dam development in Australia. At an even larger scale, Wisser et al. (2010) modelled the hydrological impact of RWH on river basins at the global scale and suggested that, while this would increase overall crop production by 35%, it would on average reduce annual flows by 18%, but with a reduction of up to 80% occurring in a single river basin. Interestingly, most of these studies concentrated on surface water impacts and ignored groundwater impacts as they were not considered relevant.

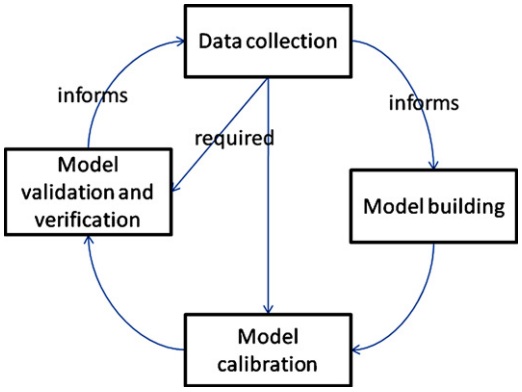


Fig. 5. The modelling and data collection cycle. Data are needed for modelling, but models can inform subsequent data collection.

Given the plethora of existing models, the question arises which specific modelling tools are most suitable or need to be developed to specifically assess the hydrological impact of RWH on both surface water and groundwater, particularly given the lack of detailed field data. Importantly, without some investment in field data, modelling cannot be the solution for assessing watershed-scale hydrological impacts of RWH. Hence, field data collection and modelling need to be developed in tandem (Silberstein, 2006) (Fig. 5).

4. Discussion

It is clear that artificial recharge using RWH is seen as a possible solution to decline of groundwater tables in India (Kendy and Bredehoeft, 2006; Sophocleous, 2000). While some of the reviewed literature suggests RWH is a useful method of replenishing aquifers, quantification of RWH recharge impact has often focussed on local, small-scale hydrological effects (e.g. Srivastava et al., 2009). Most of the studies have not considered larger watershed hydrological impacts, such as downstream trade-offs or surface water–groundwater interactions, where the use of one type of water affects the availability of the other (Badiger et al., 2002; Sharda et al., 2006).

Despite the conventional wisdom that RWH has positive impacts for irrigated agriculture by locally increasing groundwater availability, the preceding review highlights there is a serious lack of quantification of the overall watershed water balance and groundwater sustainability impacts of RWH, and this has been acknowledged in many papers (Barah, 1996; Glendenning and Vervoort, 2010, 2011; Kumar et al., 2006, 2008; Sakthivadivel, 2007; Shah, 2007). This means that there is an urgent need for improved understanding of how RWH functions and what impacts

RWH structures have on groundwater availability, as well as on the local and downstream environment (Gale, 2005). In addition, modelling studies examining RWH hydrological impacts in India have generally lacked coupled groundwater and surface water impacts (e.g. Gore et al., 1998; Tripathi et al., 2005), or have lacked sufficient data to thoroughly explore specific watershed impacts (Glendenning and Vervoort, 2011). To identify where progress in RWH impact studies can be made, this next section explores the documented impacts of RWH and identifies the research gaps.

4.1. The documented hydrological impacts of RWH at multiple scales and the research gaps

As has been highlighted earlier, the general public view of the local hydrological impacts of RWH for recharge is that this is positive and results in increased crop production (e.g. Srivastava et al., 2009). However, it is also clear that the cumulative hydrological impact of RWH, which are essentially small farm dams, on streamflow can be significant (Calder et al., 2008). In addition, land use changes, such as the development of more irrigated agriculture can be the result of perceived or real groundwater increases as a result of RWH (Calder et al., 2008; Sharma, 2002). This change in land use would cause feedback such as a net decrease in the amount of locally available water on top of decreased availability of water for downstream users (Batchelor et al., 2002; Glendenning and Vervoort, 2011). Other examples of land use changes aimed at delaying or storing water in the soil such as conservation tillage and forestation have also been shown to decrease streamflow (McVicar et al., 2007; Wang et al., 2009) (Table 1).

Spatial variability of rainfall can impact the local and watershed scale hydrological impact of RWH (Glendenning and Vervoort, 2010) (see Fig. 6). The effect of rainfall variability on the hydrological impact of RWH (both locally and at the watershed scale) needs to be quantified and long-term implications assessed. Along the same lines, a more detailed quantification of the suggested lateral flows (Glendenning and Vervoort, 2010) and the possible positive impact of such lateral flows on downstream water users also needs further research.

This also indicates that research is needed on the specific climate, hydrogeology, landform, and RWH structure combinations that make RWH for recharge a worthwhile watershed investment (Fig. 6). While some of this has been addressed in past GIS based studies (Section 3.3), we believe that these did not cover the complexity of local and watershed considerations highlighted in this review. We hypothesise that the local recharge impact of the structures would increase with increased local vertical transmissivities (but low horizontal transmissivities) and with more spatially homogeneous, but temporally more variable rainfall distributions. In addition, we can hypothesise that there would be an optimal size and number of structures in a watershed (Calder et al., 2008). Finally, if the watershed background recharge rates are similar to those from the RWH structures, then building the structures would not be very cost effective.

4.2. Other research gaps related to impacts of RWH beyond the water balance

Apart from a recent study by Wani et al. (2011), there is limited scientific work in India to highlight any positive effects of RWH on general natural resources. Conceptually, slowing down the flow of water within a watershed should result in increased opportunities for water filtering, ecosystem water use and a general shift towards green water use (Wisser et al., 2010). This is also related to the opportunity for increased erosion control and a general greater local awareness of natural resource management, which can be captured by the term "ecosystem services" (Raju et al., 2009). Apart from one social sciences paper (Cochran and Ray, 2009) and some sideways remarks (Bouma et al., 2011), we found little quantification of these aspects. This despite the fact that the origins of the watershed development program in India focussed on erosion control (Raju et al., 2009).

There is also very little research on the continued management of RWH structures, the timeframes for silt build-up and the decrease in recharge potential over time, and how management of silt layers might affect this. Furthermore, the quality of the captured water is rarely discussed, nor is the potential for human and animal disease propagation (Agarwal and Narain, 1997).

Another possible impact, though remote, is that changes in the green water profile in a watershed through increase in vegetative cover, could impact the local climate (e.g. Seneviratne et al., 2010). Combined efforts in forest management, RWH and increased crop production could alter the overall albedo, latent heat and water cycles affecting the local climate.

Finally, much of the economic research examines watershed development programs impact on welfare and calculates cost–benefit analyses, where RWH is one of many soil and water conservation measures (Joshi et al., 2005, 2008). Changes in externalities such as environmental services due to increased vegetation and or other benefits are generally not considered. Of course, negative externalities, such as changes in the salinity profile or river flows should also be considered.

4.3. The way forward

4.3.1. Absence of data

From Sections 3.2 and 4.1, it is clear that data to drive suitable models is a major limitation in assessing the regional and watershed scale hydrological impact of RWH. One problem is that the required data is sparse, particularly in semi-arid India where RWH seems most suitable. The second problem is that measuring hydrological impacts of RWH is not simple, as discussed in Section 3, as determining recharge and the impact on groundwater tables beyond local scales is complex (Fig. 3). Additional data would aid in the calibration and validation of water balance models to extrapolate RWH impacts to a larger watershed scale (Fig. 5).

Extensive data collection through hydrological instrumentation while beneficial, might not be feasible given the cost of, and required capacity to operate such equipment. Fortunately, the recent roll-out of hydrometric stations across India promises more data in the future (Kumar et al., 2005).

In terms of current strategies, there are a few options. Linking RWH assessments of hydrological impact with prediction of flow in ungauged basins (PUB) could expand the number of watersheds that can be assessed (Sivapalan et al., 2003). This involves taking rainfall runoff models that have been calibrated on gauged watersheds and applying it to ungauged watersheds with similar characteristics. Using annual data, the application of regionalised models for predicting flow in ungauged basins has been well tested in India and other countries (Castellarin et al., 2007).

Three further strategies can be considered. Access to summarised datasets of RWH research in a national database combined with descriptive metadata could further improve model development and calibration. The use of remote sensing and other novel techniques for calibration of hydrological models is also an encouraging step (Immerzeel and Droogers, 2008; Immerzeel et al., 2008). Finally, statistical techniques such as pedotransfer functions can further play an important role in predicting variables where field data is otherwise absent (Adhikary et al., 2008).

4.3.2. Suitability of models

Having addressed the data problem, the next question is whether current models can actually assess hydrological impacts

Table 1
Summary of documented impacts and research gaps of RWH for groundwater recharge at the local and watershed scale.

	Local scale	Watershed scale
Documented impacts	- Local increased groundwater availability for irrigation users (i.e. Badiger et al., 2002; Glendenning and Vervoort, 2010) - Increased local crop production (Srivastava et al., 2009; Wisser et al., 2010) - Lack of demand control and increases in irrigated crop production (Calder et al., 2008; Glendenning and Vervoort, 2011) - High evaporation due to high surface to depth ratio of storages (Neumann et al., 2004) - Low efficiency of RWH in some areas (Sharda et al., 2006)	- Increased green water and increased production (Bouma et al., 2011) - Decrease in blue water for downstream ecology and human uses similar to farm dams (Schreider et al., 2002) - Overall decrease in blue water due to changes in land use (Sharma and Thakur, 2007) - Investment does not increase watershed welfare from a production perspective (Bouma et al., 2011)
Research gaps, hydrological	- Can RWH create fresh groundwater layers over underlying saline water (Asghar et al., 2002)? - Do decreases of overland flow velocities and sediment entrainment result in better erosion control (Bouma et al., 2007)? - How do we quantify the lack of control of stored groundwater (loss of water through lateral flows) (Glendenning and Vervoort, 2010)? - What are the detrimental impact on downstream soil and water salinity?	- Decreased flow velocity resulting in ecological benefits for groundwater dependent systems and overall natural resource management. - Upstream RWH might lead to increased groundwater availability downstream due to lateral flows. - Are there increased salinity risks in hydrogeologically sensitive areas (George et al., 1997)? - What are the general impacts of RWH on water quality and human health related water issues? - How does RWH affect externalities such as related to environmental flows and other ecosystem services (Raju et al., 2009)?
Research gaps, socio-economic	- Impact of managing maintenance – collective action (Bouma et al., 2007). RWH might lead to possible increased local awareness and participation in water resource management if RWH is NGO driven (Bouma et al., 2007) - Suggested poverty alleviation is questionable (Bouma et al., 2011) - Does RWH management offer increased social cohesion (Cochran and Ray, 2009)?	- How can RWH contribute to collective at the watershed scale? Possible increased local awareness and participation in water resource management if RWH is NGO driven (Bouma et al., 2007) - Does RWH management offer increased social cohesion (this could also apply to the watershed scale) (Cochran and Ray, 2009)?

of RWH. For examples, models might be used to look at watershed-scale hydrological impacts of adding RWH structures in a watershed by running scenarios with and without RWH structures, and these can be used for follow-up socio-economic analysis (e.g. Garg et al., 2011).

The review has highlighted that most reported modelling approaches do not provide accurate impact analysis due to weaknesses in the modelling framework. Different hydrological processes dominate different land uses and climatic conditions, and an appropriate model that best represents the problem and

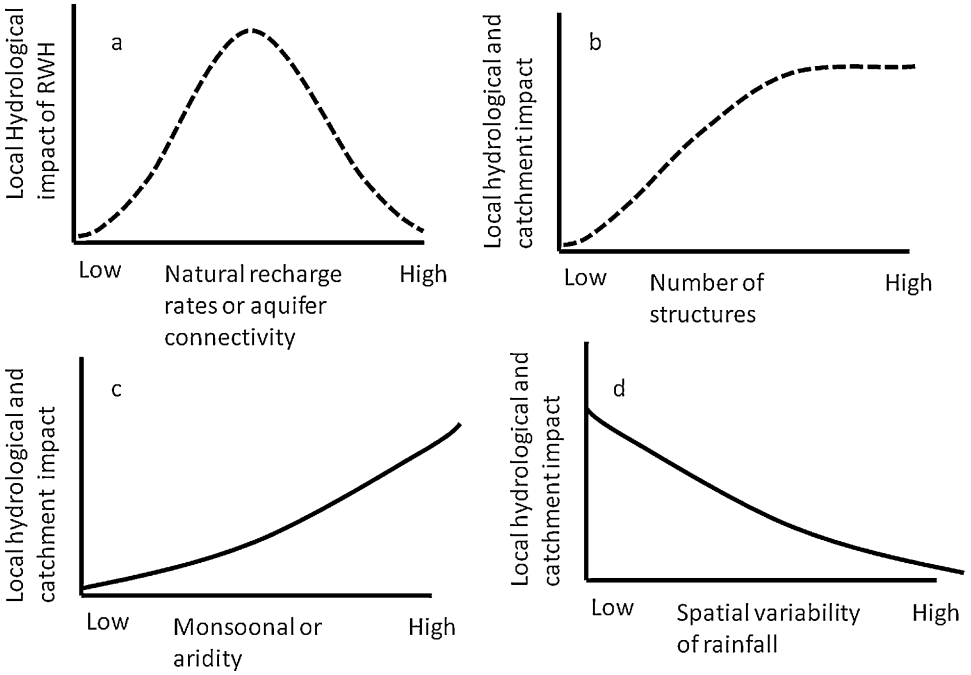


Fig. 6. Conceptual relationships indicating relative trade-offs to consider for the development of RWH in a watershed: (a) the effect of natural recharge or aquifer conductivity; (b) the effect of the number of structures; (c) the impact of rainfall temporal variability; and (d) the effect of the spatial variability of rainfall.

the landscape must be chosen. For example, in the case of RWH for groundwater recharge, runoff prediction and the description of attributes of the soil surface might be more important than other factors. This is because the areas, where this type of RWH occurs, typically receive high intensity storm rainfall events in the monsoon. Therefore, a model based on daily time steps would be better suited for the estimation of recharge, because recharge is generally a larger component of the water budget at daily scales. Smaller time steps can also minimise accumulation errors from water balance models (Scanlon et al., 2002), so can give reasonable estimates of recharge because they generally work on time series that are shorter than a year (Walker and Zhang, 2002).

A key element for models assessing the long-term hydrological impacts of RWH is whether they can accommodate major shocks to the system, such as drought and major land use change. This is particularly important for hydrological models that are subsequently used to examine socio-economic scenarios (e.g. Bouma et al., 2007, 2011). Given that varied land uses would have to be modelled to assess the hydrological impact of RWH, including spatial variability in the model framework would be important. However, every model choice needs to be accompanied by a clear objective combined with an understanding of the key processes and data appropriate to achieve the objective (Finch, 1998). With increased complexity more parameters are required and this increases data collection effort, but also the potential for uncertainty (Beven, 2001). More importantly, equifinality in modelling means that different model structures can give similar calibration results (Beven, 2002) and thus clear choices between models can be difficult. Better understanding of model calibration and validation needs to be incorporated with understanding of non-stationarity concepts (Milly et al., 2008; Scanlon et al., 2002; Silberstein, 2006).

As a result, we can identify base criteria for a watershed model to study the impact of RWH at multiple scales:

1. It needs to contain programming code that can be adjusted to accommodate RWH structures and other necessary local modifications.
2. It needs to describe groundwater surface water interaction in a physically plausible way.
3. It should be at least semi-distributed to accommodate variation in landuses.
4. Calibration should be based on multiple criteria to counter equifinality problems.

Of all models reviewed, we believe the SWAT is an interesting candidate for further development. It is an open source water balance model that has been regularly applied in India (Gosain and Rao, 2004; Kelkar et al., 2008; Tripathi et al., 2005). Table 2 specifically highlights a number of studies that have used SWAT to analyse the benefits and potentials of RWH, but that did not yet explicitly model the hydrological impacts of RWH.

An advantage of SWAT is that it can work quite well with satellite derived data mentioned in Section 3, and, in particular, it can be used for predictions in ungauged basins (Immerzeel and Droogers, 2008). Furthermore, being open source, the SWAT code can easily be modified to either add new processes or modify existing processes in order to tailor the model for the particular use or address deficiencies (e.g. Ouessar et al., 2009). However, the open source nature of SWAT can result in less well tested or underdeveloped programming code and hence there are currently still a number of areas which require further testing and development, such as the routing routines (Gassman et al., 2007), the conceptual description of the groundwater surface water interaction and the lack of a physically realistic description of the groundwater table at the hydrological response unit (Ouessar et al., 2009).

4.4. Possible criteria for assessing the design of watershed development using RWH for recharge

Based on our review, we propose criteria that would need to be considered before developing and expanding RWH, specifically for increasing recharge in a watershed. The criteria combine hydrological impact considerations at multiple scales that arise from this review combined with our understanding of the socio-economical issues that we identified.

Considering the local aspects of RWH, the main suggested benefits (as listed in Table 2) increased groundwater availability for agricultural irrigation, which could lead to increase in crop yield, and the potential to increase community awareness about water use and the environment (Kumar and Kandpal, 2003). The negative impacts of RWH mainly centre on lack of control of the captured water and downstream effects and the under-researched area of water quality and human health. Summarising, for RWH to provide successful increases in groundwater availability at a single location we need locations with:

- strongly variable rainfall systems in a temporal sense, such as monsoonal systems, but with limited spatial variability (Fig. 6);
- local high infiltration rates relative to the surrounding landscape;
- a substantial watershed area to accumulate sufficient captured water to a reasonable depth so evaporation losses can be limited;
- favourable anisotropy in the aquifer with $K_y < K_z$, i.e. the lateral transmission (K_y) is smaller than the vertical transmission (K_z) and not all the infiltrating water is lost to lateral flow (Figs. 3 and 5). This is particularly important for aquifers with high salinity to allow the development of a fresh water lens on top of saline groundwater;
- preferably shallow alluvial aquifers rather than fractured hard-rock aquifers, as the former are better for shallow hand-dug wells (i.e. poorer farmers). In this case the low cost of development and construction will balance the increased irrigation potential and possible poverty alleviation gains;
- a low permeability layer below the productive aquifer, so vertical leakage is minimised; and
- a strong community institution to guarantee maintenance of the structures, such as removing the siltation layer which would decrease the recharge potential.

Considering the watershed scale aspects of RWH, the benefits can be (as listed in Table 2) increased ecological resilience, and increases in cropped irrigation area and production. However the negatives include upstream downstream inequality (i.e. Bouma et al., 2011) and possible unquantified salinity risks in regions where such risks exist. Therefore for RWH to provide successful increases in groundwater availability at the watershed scale we need, in addition to the local criteria:

- opportunities for RWH upstream and downstream to alleviate inequality and loss of downstream blue water, or high lateral transmissivities (in contrast to 4 in the local criteria above) to increase groundwater downstream;
- opportunities to manage overall water demand by increasing water use efficiency in irrigation;
- planning to limit the total number of structures in the watershed, which will depend on the size of the watershed (Fig. 6), as too many structures decrease the overall recharge potential;
- relatively low background diffuse recharge, but sufficient locations for successful local RWH;
- low salinity hydrogeology if lateral flows are high; and
- overall relatively flat watershed topography, to limit lateral flows beyond the watershed, but with local elevation differences.

Table 2
Examples of modelling studies using SWAT related to benefits and potentials of general RWH management.

Region	Description	Reference
Thukela River Basin, South Africa	Estimates water available for in situ water harvesting and supplemental water demand	Andersson et al. (2009)
Lakwar watershed, India	Identified rainwater harvesting as being feasible (household tanks)	Kelkar et al. (2008)
Jeffara, wadi Koutine, Tunesia	Developed SWAT-WH (SWAT-Water Harvesting) for use with water harvesting structures – recommends linking SWAT-WH to a grid based routing model	Ouessar et al. (2008)
Doddahalla watershed, Bijapur, India	Identified areas where rainwater harvesting can be placed	Gosain and Rao (2004)

A final point of consideration is the landscape scale aspects of RWH (interconnected watersheds in a river basin). At this point socio-economic and development aspects at a much larger scale become important, but these remain largely unquantified. In particular:

- the need to weigh up poverty alleviation and local socio-economic development against the reduction of blue water for irrigation development and users further downstream, such as demonstrated in the Krishna Basin (Garg et al., 2011);
- other opportunities for ecological and hydrological restoration (such as reforestation and erosion management) in combination with RWH;
- the balance between government control on groundwater relative to local control of groundwater; and
- the trade-off between ecological and economic assets at a river basin scale.

5. Conclusions

The benefits of RWH for increased groundwater recharge remain a point of discussion in the literature. In particular, the complexities associated with understanding and measuring groundwater recharge have made it very difficult to quantify the hydrological impacts of RWH at the local and watershed scales. This has also made it difficult to quantify any negative impacts associated with RWH for recharge. However, there are a number of new research avenues that may greatly assist in clarifying the hydrological impacts of RWH. Importantly, these options do not necessarily have to be expensive. Open source software and freely available datasets, such as those derived from satellite images, gridded rainfall datasets and soil data, have been successfully used to model the hydrology globally.

However, modelling without further data collection would not lead to further insights.

There remain a large number of research gaps related to the hydrological and social impacts of RWH for recharge. Continued and new research into the areas identified in this review would greatly assist further policy development. We have highlighted a major lack of quantifiable hydrological studies at the watershed scale and also a lack of local scale data. While models are powerful and will continue to play a role in management of RWH, further model development and integration with new data sources is needed. There is very little understanding of the impact of siltation on the RWH structure hydrology and how siltation may be managed. Aspects of water quality and the interplay between RWH for recharge and groundwater salinity are also lacking from the reviewed research. Furthermore, there is a lack of socio-economic studies on externalities of RWH for recharge and a lack of social studies examining problems with local RWH construction and maintenance.

Finally, based on the reviewed literature, we propose assessment criteria that could be used to evaluate development of current and future RWH for recharge. These criteria take into account the reviewed biophysical aspects as well as some of the identified socio-economic aspects at a range of spatial scales.

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